Unit: 6 Unit 6: Systems of Linear Equations and Inequalities **Name:** ________________
Topic: Systems of Linear Inequalities by Graphing **Date:** __________

Advanced (Habanero)

Q1. What is the x-intercept of the line $2x + 3ygeq6$?
(A) -3
(B) 0
(C) 3
(D) -2
(E) 2

Q2. Which inequality has the solution region below the line $y = x + 1$?
(A) $y < x + 1$
(B) $y > x + 1$
(C) $yleqx + 1$
(D) $ygeqx + 1$
(E) $y = x + 1$

Q3. The system of inequalities $x + ygeq2$ and $x - yleq1$ has a solution region that is
(A) a triangle
(B) a rectangle
(C) a quadrilateral
(D) a circle
(E) a line segment

Q4. The feasible region for the system $2x+3ygeq6$ and $x-yleq1$ is
(A) unbounded
(B) bounded
(C) a single point
(D) a line segment
(E) empty

Q5. The graph of $y > 2x - 3$ has the same solution region as the graph of
(A) $ygeq2x - 3$
(B) $y < 2x - 3$
(C) $yleq2x - 3$
(D) $y = 2x - 3$
(E) $yeq2x - 3$

Q6. The system $x + 2ygeq4$ and $2x + yleq5$ has no solution if
(A) $x = 2$ and $y = 1$
(B) $x = 1$ and $y = 2$
(C) $x = 0$ and $y = 0$
(D) $x = 3$ and $y = 1$
(E) $x = 2$ and $y = 3$

Q7. The solution region of $y > x^2 - 4$ and $yleq - x + 2$ is
(A) a parabolic segment
(B) a triangular region
(C) a circular region
(D) an elliptical region
(E) a rectangular region

Q8. The system of inequalities $x+2ygeq6$ and $x-2yleq-2$ has the solution region that is
(A) a half-plane
(B) a line segment
(C) a parabolic segment
(D) a triangle
(E) a rectangle

Open Ended Questions (FRQ)

Q9. Graph the system of inequalities $ygeqx - 2$ and $yleq - x + 1$ on the same coordinate plane, and identify the vertices of the solution region.

Q10. The system of inequalities $x + ygeq4$ and $x - yleq1$ represents the feasible region for a company's production. Describe the feasible region, and find the maximum value of $x + 2y$ in this region.

Chef's Tips

- Emphasize the importance of labeling the x- and y-axes when graphing inequalities.
- Remind students to consider the non-strict inequalities and their implications on the solution regions.
- Encourage students to use graphical representations to verify algebraic solutions.